PRANA-TWIN System Architecture Diagram

Multi-Stream Data Fusion, Compartmental ODEs, Ensemble ML Predictors, and Doctor Decision Support UI

Layer 1: Multi-Stream Data Ingestion

- ▶ Static EHR Stream: Demographics, HbA1c (8.6%), eGFR, Lipids, Prescriptions
- ▶ Dynamic IoT Stream: 5-min CGM Glucose, PPG Pulse, HRV (RMSSD), Steps
- ▶ Indian Dietary Profile: High-GI staple meal carb bolus (Rice, Parathas, Sweets)
- ▶ Circadian Telemetry: Nighttime fasting epochs & autonomic sleep staging
- ▶ Real-time Validation: Quality-checked telemetry streams ingested asynchronously

Layer 2: Hybrid Digital Twin Core

- ▶ Physiological ODEs: Bergman Minimal Model (Plasma Glucose + Remote Insulin)
- ▶ Multimodal Feature Fusion: Temporal derivative extractor (dG/dt, d2G/dt2)
- ▶ Forward Horizon Regressor: Gradient Boosted 24-step trajectory (0-120 mins)
- ▶ Adverse Event Classifiers: Acute Spike (>220 mg/dL) & Nocturnal Hypo (<70 mg/dL)
- ▶ Counterfactual Sandbox Engine: Instant simulation of Diet, Walk & Meds
- ▶ Explainable AI (XAI): Dynamic attribution weights for clinical drivers

Layer 3: Clinical Decision Support UI

- ▶ Patient Triage Selector: Real-time risk status (Critical, High Risk, Moderate, Stable)
- ▶ Interactive Organ Twin: SVG avatar with glowing Pancreas, Heart, Kidneys, Liver
- ▶ 2-Hour Predictive Chart: Horizon graph with 90% confidence bands & thresholds
- ▶ Counterfactual Sliders: Dynamic carb titration, walking minutes, and medication
- ▶ RSSDI / ICMR Clinical Audit: Instant export of AGP metrics & consultation notes